

Toward Ground-Truth Optical Coherence Tomography via Three-Dimensional Unsupervised Deep Learning Processing and Data

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Abstract: In Optical Coherence Tomography, speckle noise is an unavoidable problem and one that has been met with many methods and strategies of denoising. Speckle-Modulating Optical Coherence Tomography (SM-OCT) is among the most successful denoising strategies, but it carries notable limitations.. This opens the doors for methods that can overcome these limitations seen with Speckle-Modulating Optical Coherence Tomography without having to sacrifice the valuable spatial resolution it brings. Researchers attempting to accomplish this goal developed Toward Ground-Truth Optical Coherence Tomography (tGT-OCT) [1], utilizing convolutional neural networks (CNN) in combination with Generative Adversarial Networks (GAN). A 3D convolutional teacher network learns from purely noisy data using a Noise2Noise strategy, and a 2D student network mimics its behavior through adversarial training. Using these in combination has proven successful in denoising speckle, enabling fast, high-quality, hardware-free OCT denoising that has proven to be able to generalize well across tissues and scanners, matching the level achieved by Speckle-Modulating Optical Coherence Tomography. The researchers demonstrate this through qualitative comparisons, which are best illustrated by the comparison of the two post-processed images of Toward Ground-Truth Optical Coherence Tomography and Speckle-Modulating Optical Coherence Tomography.

1. Introduction

Optical Coherence Tomography (OCT) is a powerful non-invasive imaging modality widely used in medicine, using near-infrared light to penetrate semi-transparent or scattering tissues (e.g., the retina, skin, or coronary arteries) and provides cross-sectional or 3D images of subsurface structures, able to image objects beyond their surfaces utilizing the power of low-coherence interferometry. However, there is one unavoidable obstacle when utilizing Optical Coherence Tomography, known as speckle noise.

Speckle noise results from random interference patterns produced by scattered coherent light. This speckle noise degrades image quality overall, as well as the detail and resolution of the images produced by Optical Coherence Tomography. Speckle noise is random and uncorrelated; therefore, imaging techniques such as Speckle-Modulating Optical Coherence Tomography, which utilizes a light manipulation technique, inducing different phase shifts to the light through a moving diffuser. The different phases of light hitting the object allow for different speckle patterns to affect the image, revealing structures hidden by the speckle of one phase of light but revealed by a different phase of light. By taking many B-scans at the same locations, while introducing these differing speckle patterns, we can average all the images taken, and since the speckle is what is changing, but the structure remains constant, we are able to create more defined and clear denoised images revealing the structure previously hidden by speckle noise [2]. Though we are trying to beat the limitations of Speckle-Modulation Optical Coherence Tomography, it is an important validation and comparison tool when investigating new methods like Toward Ground-Truth Optical Coherence Tomography for their abilities in denoising.

2. Methods

In tGT-OCT, the researchers needed to measure how well the imaged structures align with the neural network's learned representations. To understand this, it helps to know that a B-scan is formed by stacking multiple one-dimensional A-scans, each representing tissue reflectivity along a line. The model uses spatial relationships between A-scans to predict structure better and remove speckle. This is captured in equation (1), where ρ is the correlation between two A-scans, d is their lateral distance, and ω is the Gaussian beam waist (12 μm), which also defines the OCT system's transverse resolution. Essentially, as the lateral distance increases, A-scan correlation and structural similarity decrease.

$$\rho = \exp\left(-\frac{d^2}{\omega^2}\right) \quad (1)$$

The method used to eliminate the need for a ground truth image for training and testing is called the Noise2Noise method. This is represented by Figure 1. The original 3D volumetric data set is separated into two sub-datasets, where every odd B-scan (1st, 3rd, 5th...) is put into one sub-dataset. On the other hand, all the even B-scans (2nd, 4th, 6th...) are placed into the other sub-dataset. After this division, each sub-dataset undergoes random B-scan dropping, where either one of two consecutive B-scans is removed from the sub-dataset, or two of four randomly chosen in-line B-scans are dropped. This process ensures that the two sub-datasets will have uncorrelated speckle patterns even though they were taken from the same original volumetric dataset. Lastly, a cropping block of a predetermined size ($h, w, D/4$) is used to crop the processed sub-dataset volumetric images into smaller volumetric images, thereby producing smaller paired 3D volumetric datasets suitable for training and testing, rather than relying on a single large volume. This leaves us with paired 3D datasets, one input group of noisy data and a label group of noisy data set enabling the Noise2Noise method. In theory, if a convolutional neural network can learn to map one noisy image to another, it implicitly captures the shared structure while filtering out uncorrelated noise. Now, in reality, we will not be able to predict the noise of the label dataset starting with the input data; we can, however, minimize the difference between the two if we get as close to entirely eliminating the noise from the input image as possible. We will see how this is being done through a variety of loss functions and Generative Adversarial Networks.

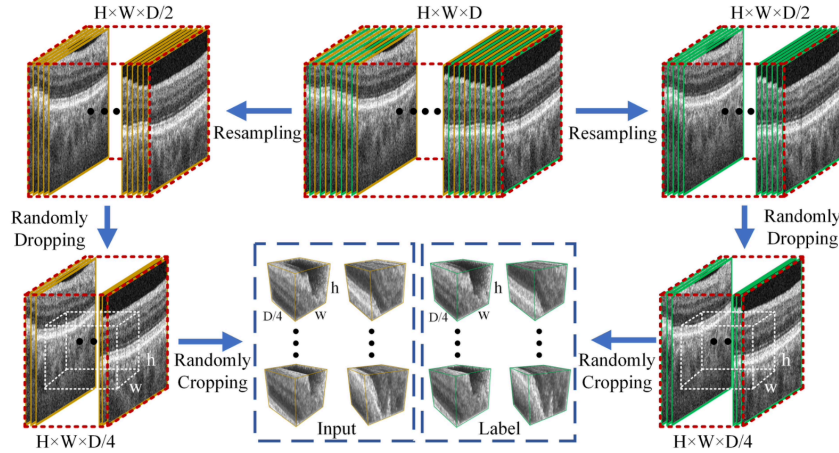


Figure 1. The method of generating paired 3D volumetric data sets. The original volumetric data is resampled into two similar volumes, which are then divided into paired volumetric blocks through random dropping and cropping.

The overview of the framework for our Toward Ground-Truth method is shown in Figure 2. The framework consists of two parts: a 3D unsupervised learning portion and a knowledge distillation portion, where a Generative Adversarial Network optimizes the convolutional neural network parameters in both the (2D) student network, acting as the Generator, and the Discriminator Network.

The 3D unsupervised learning network, also known as the (3D) Teacher Network, is a neural network made up of three parts, the first being the pre-residual layer, which is made up of a $3 \times 3 \times 3$ 3D convolutional layer with 32 filters, each performing a specific volumetric feature extraction by sliding through the input sub-datasets placed into the network in three dimensions, and following it is a $3 \times 3 \times 3$ Batch normalization layer, playing a role in stabilizing

and accelerating the training of the deep neural networks by normalizing the inputs to each layer. The second part is the residual layer, comprising 16 residual blocks, each of which consists of 6 additional layers. The layers within the residual blocks are made up of a 3D convolution layer and a normalization layer, similar to that of the pre-residual layer, but then followed by a 1x1x1 leaky ReLU layer, acting as an activation function of this neural network. These three layers are followed by an identical three layers, and a residual connection is added, allowing the neural network to focus on learning the difference (residual) between noisy and clean data, rather than trying to learn the full image from scratch, making up the formation of one complete residual block. This layer structure is then repeated 16 times, giving us a total of 16 residual blocks, which, as we stated before, make up the residual layer of the Teacher Network. Last comes the post-residual layer, where a last 3D Convolution layer, a Batch Normalization layer, and a Leaky ReLU layer are set, but this data has 32 channels, meaning it is not in a readable format; this format is only understood by the neural network, and to make it into an image, that can be made sense of, in a “readable” format, a 1x1x1 3D Convolution layer is then placed, giving us our complete 3D volumetric denoised image, the 3D volumetric Toward Ground-Truth Optical Coherence Tomography image. This is then compared against a correlated 3D volumetric image given by our label data from the sub-dataset as described in the Noise2Noise method section, using the unsupervised loss function, as described later in the paper.

Next, in the second part of our framework, the Knowledge Distillation part is made up of the Generative Adversarial Network. The Generative Adversarial Network can be further divided into two parts, one being the Student Network, as mentioned, acting as the Generator Network, and the other being a Discriminator Network. For the Student Network, it has the same format and structure as the Teacher Network, as talked about previously, the only difference being that it is handling two-dimensional data sets, single B-scans from the original 3D sub-dataset input used in the Teacher Network, rather than the input being the whole volumetric sub-dataset. After the data is processed in the Student Network and a clean 2D image is produced, it is then put into the Discriminator Network and put against a generator loss function that will adjust the neural network to optimize network parameters, as described later in the paper. This is then put into the Discriminator Network that has a similar structure as the residual portion of the Student Network enabling data processing in the network, and this is followed by a flattened layer, a dense layer, and a sigmoid function that were used to determine the classification probability, and put against a discriminator loss function to optimize network parameters, also described later in the paper.

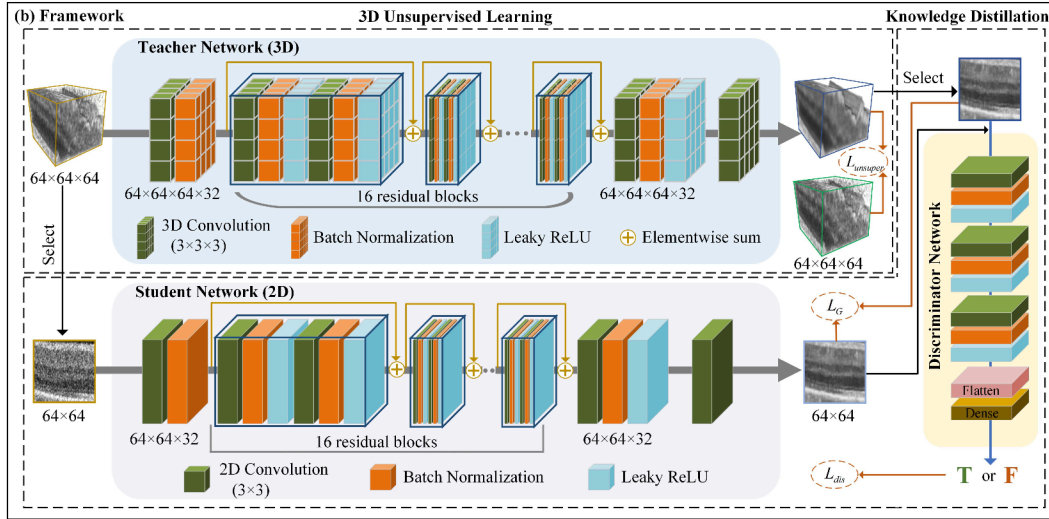


Figure 2. The Toward Ground-Truth Optical Coherence Tomography Framework. (a) The 3D Unsupervised Learning and (b) The Knowledge Distillation

There are three main loss functions used in the framework. The first is the unsupervised loss function, equation (2), which is utilized to minimize the differences between the 3D volumetric image product of the Teacher Network and the correlated label 3D volumetric image, thereby minimizing the difference between the two and successfully implementing the Noise2Noise method previously described.

$$L_{unsuper} = \frac{1}{HWF} \sum_{i,j,k=1}^{H,W,F} \|v_{i,j,k} - p_{i,j,k}\|^2 \quad (2)$$

$v_{i,j}$ = Noisy label image, $p_{i,j}$ = Denoised 3D network output image

Minimizing this function to its highest degree will ensure that we have denoised successfully, as the difference between the denoised image and the noisy label data will be minimized, which will indirectly indicate that the neural network has successfully learned the structures in the object being imaged and the noise. The next function used is the generator loss function, written as equation (3), where $G(x_{i,j})$ is the generated image from the 2D network. This loss function which works to correct the generator if it is not passing the discriminator, passing as a “real” picture, sending the loss function to negative infinity, and also to communicate to the neural network if parameters need to be changed if the discriminator labels the generators image as “fake”, which sends the loss function to 0. The last loss function is one used for the discriminator loss function, written as equation (4), similar to equation (3) this is to correct the discriminator network by maximizing the loss function if the discriminator finds a generated image is “real” or if the discriminator network finds a real image is “fake”.

$$\min_G Loss_{adv}(G) = \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(1 - D(G(x_{i,j}))) \quad (3)$$

$$\min_D Loss_{dis}(D) = -\left\{ \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(D(y_{i,j})) + \frac{1}{HW} \sum_{i,j=1}^{H,W} \log(1 - D(G(x_{i,j}))) \right\} \quad (4)$$

$x_{i,j}$ = Generated student network image, $y_{i,j}$ = Real teacher network image

3. Results and Discussion

Figure 3 shows the comparisons of how well our tGT-OCT method performs against the state-of-the-art denoising method, SM-OCT. tGT-OCT achieves a comparable level of denoising, clearly revealing structures and features similar to SM-OCT, which would not be clear or differentiable using the original noisy image. Figure 4 shows the comparison of the teacher network output image (a B-scan from it) and the student network output image, showing that they both reach comparable levels of denoising, but with a greater image contrast and more discernible blood vessel details in the teacher network compared to those of the student network. In (b3), Detailed information on the sweat duct in the epidermis and epidermal junction can be clearly observed. This shows that the 2D network of tGT-OCT is still a powerful and reliable tool, though the 3D network still produces slightly higher contrast and structural detail.

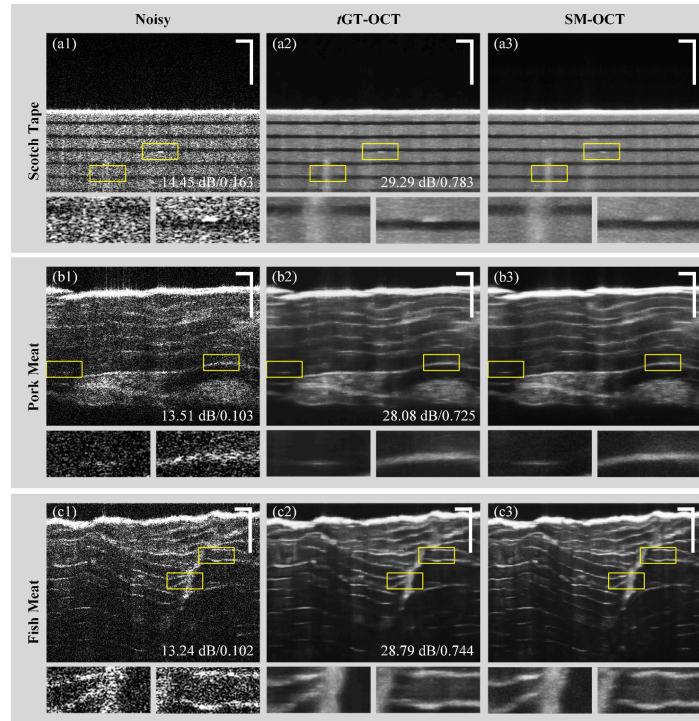


Figure 3. tGT-OCT comparison with SM-OCT and the original Noisy image. The evaluation metrics format is PSNR/SSIM. Scale bar: 200µm.

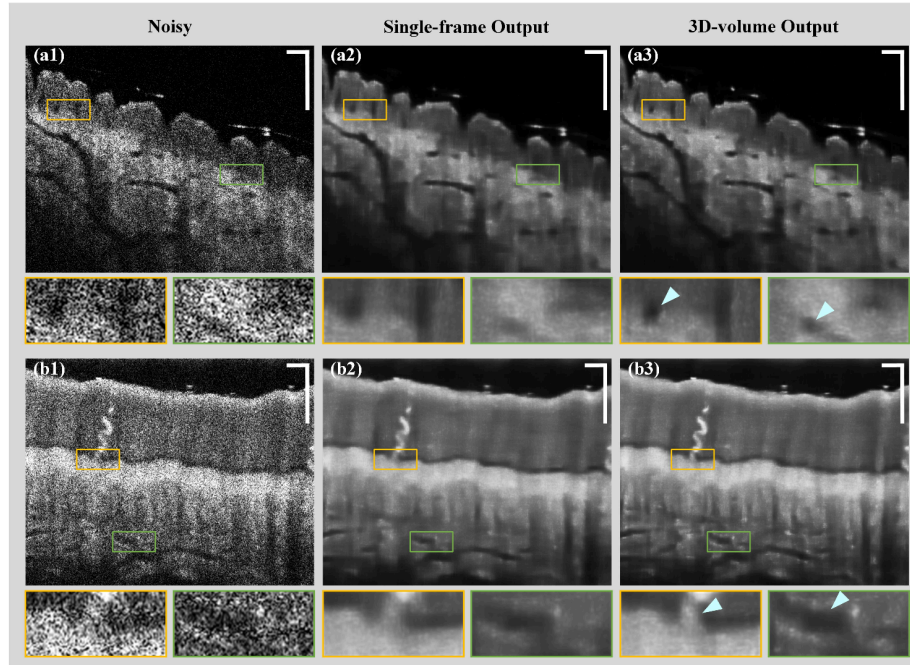


Figure 4. tGT-OCT's 3D-volume output image comparison with Single-frame Output image and the original Noisy image. Scale bar: 200 μ m.

4. Conclusion

In this paper, we examined a novel approach for denoising OCT images without the need for clean ground truth data. The method, tGT-OCT, leverages the uncorrelated nature of speckle noise between alternating B-scans to train a 3D convolutional teacher network using an unsupervised Noise2Noise loss. By understanding the spatial relationship between A-scans, quantified by their correlation as a function of lateral distance and the system's Gaussian beam waist, the model learns to identify and preserve true anatomical structures while suppressing noise. The output of this volumetric teacher network is then used to guide a 2D student network, which replicates the denoised results more efficiently. A discriminator network further refines the student's performance by encouraging perceptually realistic outputs. Together, this architecture enables a real-time, more practical, hardware-free speckle reduction alternative and generalizes well across different sample types. The approach not only improves image clarity but also maintains essential features critical for diagnostic interpretation, offering a scalable and accessible alternative to hardware-based denoising solutions.

5. References

- [1] Ni G, Wu R, Zheng F, Li M, Huang S, Ge X, Liu L, Liu Y. Toward Ground-Truth Optical Coherence Tomography via Three-Dimensional Unsupervised Deep Learning Processing and Data. *IEEE Trans Med Imaging*. 2024 Jun;43(6):2395-2407. doi: 10.1109/TMI.2024.3363416. Epub 2024 Jun 3. PMID: 38324426.
- [2] Liba, O., Lew, M., SoRelle, E. et al. Speckle-modulating optical coherence tomography in living mice and humans. *Nat Commun* 8, 15845 (2017). <https://doi.org/10.1038/ncomms15845>.